Chihiro Taguchi

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Education

08/2022 – Present	**University of Notre Dame** PhD Candidate in Computer Science and Engineering. Expected graduation in 05/2027.	IN, United States
09/2021 – 08/2022	**University of Edinburgh** Master of Science by Research (MScR) in Linguistics with Distinction.	Edinburgh, United Kingdom
10/2020 – 09/2022	**Nara Institute of Science and Technology** Master of Engineering.	Nara, Japan
09/2018 – 06/2019	**School of Oriental and African Studies, University of London** Exchange program.	London, United Kingdom
04/2015 – 09/2019	**Keio University** Bachelor of Laws (LL.B.).	Tokyo, Japan

Training

07/2025 – 08/2025	**Linguistic Society of America (LSA) Institute**	Eugene, OR, United States
06/2024 – 10/2025	**Leadership Advancing Socially Engaged Research (LASER)**	University of Notre Dame
06/2023 – 07/2023	**Linguistic Society of America (LSA) Institute**	Amherst, MA, United States
05/2023 – 06/2023	**Summer Language Abroad Program (Quechua)**, University of Notre Dame	Quito, Ecuador
02/2018	**3rd Miyako Language Documentation Workshop**	Okinawa, Japan
07/2016 – 08/2016	**ARMACAD Armenian Studies Summer School**	Yerevan, Armenia

Experience

03/2026 – 07/2026	**Intern** at Sakana AI (Tokyo, Japan).
08/2025 – 12/2025	**Teaching Assistant** for CSE 40657 “Natural Language Processing”.
05/2025 – 09/2025	**Intern** at Sakana AI (Tokyo, Japan).
01/2025 – 04/2025	**Intern** at Megagon Labs (Mountain View, CA, USA).
04/2021 – 09/2021	**Research Assistant** at the National Institute of Informatics (Tokyo, Japan).

Peer-reviewed Publications

2027 **Chihiro Taguchi**. “MaxEnt Modeling of Verbal Honorific Marking in Early Middle Japanese: Evidence for an Actor–Undergoer-Based Analysis.” Accepted at the Annual Meeting of the Linguistic Society of America.

2026 **Chihiro Taguchi**, Éric Le Ferrand, Hirosi Nakagawa, Hitomi Ono, Kanji Kato, Emily Prud’hommeaux, David Chiang. “Pretrained self-supervised speech models can recognize unseen consonants.” Interspeech. https://www.isca-archive.org/interspeech_2026/taguchi26_interspeech.html

2026 Yotaro Kubo, Richard Sproat, **Chihiro Taguchi**, Llion Jones. "Building Tailored Speech Recognizers for Japanese Speaking Assessment." Interspeech. https://www.isca-archive.org/interspeech_2026/kubo26_interspeech.html

2026 **Chihiro Taguchi**, Richard Sproat. "Creating ConLangs to Probe the Metalinguistic Grammatical Knowledge of LLMs." ACL. https://aclanthology.org/2026.findings-acl.1455

2026 **Chihiro Taguchi**, Yukinori Takubo, David Chiang. "Automatic Speech Recognition for Documenting Endangered Languages: Case Study of Ikema Miyakoan." LREC. https://aclanthology.org/2026.lrec-1.284

2026 **Chihiro Taguchi**. "Notre Dame NLP at Mozilla Common Voice Spontaneous Speech ASR Shared Task: Language-agnostic augmentation of spontaneous speech data." In submission.

2025 **Chihiro Taguchi**, Richard Sproat. IASC: Interactive Agentic System for ConLangs. arXiv preprint. https://arxiv.org/pdf/2510.07591v1

2025 **Chihiro Taguchi**, Seiji Maekawa, Nikita Bhutani. "Efficient Context Selection for Long-Context QA: No Tuning, No Iteration, Just Adaptive-*k*." EMNLP Main. https://aclanthology.org/2025.emnlp-main.1017

2025 **Chihiro Taguchi**, Seng Mai, Keita Kurabe, Yusuke Sakai, Georgina Agyei, Soudabeh Eslami, David Chiang. "Languages Still Left Behind: Toward a Better Multilingual Machine Translation Benchmark." EMNLP Main. https://aclanthology.org/2025.emnlp-main.1018

2025 Hiroyuki Deguchi, Go Kamoda, Yusuke Matsushita, **Chihiro Taguchi**, Kohei Suenaga, Masaki Waga, Sho Yokoi. "SoftMatcha: A soft and fast pattern matcher for billion-scale corpus searches." ICLR. https://arxiv.org/abs/2503.03703

2024 **Chihiro Taguchi**, David Chiang. "Language Complexity and Speech Recognition Accuracy: Orthographic Complexity Hurts, Phonological Complexity Doesn't." ACL Main. **Outstanding Paper Award**, **Senior Area Chair's Award**. https://aclanthology.org/2024.acl-long.827

2024 Yuhi Matogawa, Yusuke Sakai, Taro Watanabe, **Chihiro Taguchi**. "Japanese Rule-based Grapheme-to-phoneme Conversion System and Multilingual Named Entity Dataset with International Phonetic Alphabet." SIGMORPHON at NAACL. https://aclanthology.org/2024.sigmorphon-1.9

2024 **Chihiro Taguchi**, Jefferson Saransig, Dayana Velásquez, David Chiang. "Killkan: The Automatic Speech Recognition Dataset for Kichwa with Morphosyntactic Information." LREC-COLING. https://aclanthology.org/2024.lrec-main.852

2024 Tatsuya Aoyama, **Chihiro Taguchi**, Nathan Schneider. "J-SNACS: Adposition and Case Supersenses for Japanese Joshi." LREC-COLING. https://aclanthology.org/2024.lrec-main.839

2024 Jonathan Washington, Çağrı Çöltekin, Furkan Akkurt, Bermet Chontaeva, Soudabeh Eslami, Gulnura Jumalieva, Aida Kasieva, Aslı Kuzgun, Büşra Marşan, **Chihiro Taguchi**. "Strategies for the Annotation of Pronominalised Locatives in Turkic Universal Dependency Treebanks." MWE-UD at LREC-COLING. https://aclanthology.org/2024.mwe-1.25

2024 **Chihiro Taguchi**, Jefferson Saransig. "Non-discourse-configurationality in Imbabura Kichwa." In the *Proceedings of the Linguistic Society of America*. https://journals.linguisticsociety.org/proceedings/index.php/PLSA/article/view/5687

2023 **Chihiro Taguchi**, Yusuke Sakai, Parisa Haghani, David Chiang. "Universal Automatic Phonetic Transcription into the International Phonetic Alphabet." INTERSPEECH. https://www.isca-archive.org/interspeech_2023/taguchi23_interspeech.html

2023 **Chihiro Taguchi**, David Chiang. "Introducing Morphology in Universal Dependencies Japanese." 6th Universal Dependencies Workshop. https://aclanthology.org/2023.udw-1.8

2022 **Chihiro Taguchi**. "Mermaid Construction in Lexical Functional Grammar." In the *Proceedings of LFG'22*. https://lfg-proceedings.org/lfg/index.php/main/article/view/19

2022 **Chihiro Taguchi**. "Deconstructing Mermaid Constructions: From Control and Raising in Tatar." Research Master's Thesis (Distinction), University of Edinburgh.

2022 **Chihiro Taguchi**. "Natural Language Processing for Lower-Resource Code-Switching Languages: Case Studies on Transliteration and Resource Building for Tatar." Master's Thesis, Nara Institute of Science and Technology.

2022 **Chihiro Taguchi**, Sei Iwata, Taro Watanabe. "Universal Dependencies Treebank for Tatar: Incorporating Intra-Word Code-Switching Information." EURALI at LREC. https://aclanthology.org/2022.eurali-1.17

2021 **Chihiro Taguchi**, Yusuke Sakai, Taro Watanabe. "Transliteration for Low-Resource Code-Switching Texts: Building an Automatic Cyrillic-to-Latin Converter for Tatar." CALCS at NAACL. https://aclanthology.org/2021.calcs-1.18

Book chapters

2027 Chapter "Grammaticalization of modal nominal predicates in Tatar" in *Studies in Grammaticalization in Turkic*. Oxford University Press.

2027 Chapter "Universal Automatic Phonetic Transcription" in *The Transcription of Atypical Speech: A Book of Readings*. University of Toronto Press.

Presentations

03/2027 **Chihiro Taguchi**, Kiyono Fujinaga-Gordon, Akitaka Yamada. "Technological optimism, skepticism, and apathy: What the Miyakoan-speaking community want from AI". Accepted as a presentation at te 11th International Conference on Language Documentation & Preservation (ICLDC) at the University of Hawai'i at Mānoa (peer-reviewed).

03/2026 **Chihiro Taguchi**, "低資源言語音声認識における n-gram 言語モデルの有効性と無効性" [The (in)effectiveness of n-gram language models in low-resource speech recognition]. Poster presentation at the 32nd Annual Meeting of the Association for Natural Langugae Processing.

03/2025 **Chihiro Taguchi**, Keita Kurabe, Yusuke Sakai, Rita Seng Mai Nbanpa, "フィールドワークデータによるジンポー語機械翻訳" [Machine translation for Jinghpaw with fieldwork data]. Poster presentation at the 31st Annual Meeting of the Association for Natural Language Processing.

03/2025 "Digital documentation for diasporic data: challenges, opportunities, and solutions for working with diaspora communities." Oral presentation at the 9th International Conference on Language Documentation & Conservation (ICLDC) at the University of Hawai'i at Mānoa (peer-reviewed).

06/2024 "Killkan: The Automatic Speech Recognition Dataset for Kichwa with Morphosyntactic Information." Poster presentation at the AmericasNLP workshop at NAACL.

04/2024 "Kichwa Meets Language Technologies: Automatic Speech Recognition for Kichwa." Oral presentation at the Quechua Alliance Annual Meeting at Northwestern University (peer-reviewed).

01/2024 "El conjunto de datos de reconocimiento automático de voz para kichwa." Oral presentation at the Taller: Tecnologías Digitales y Lenguas Indígenas at the Universidad de Chile.

11/2023 "Reconocimiento automático del habla de kichwa para su inclusión y empoderamiento en el ciberespacio." Oral presentation at the VI Seminario Internacional Revitalizando Ando at the Universidad Politécnica Salesiana, Quito.

09/2023 "UD-Tatar NMCTT Treebank: Issues in annotation across Turkic UD." Oral presentation at the UD Turkic Workshop.

08/2023 "Grammaticalization of modal nominal predicates in Tatar." Oral presentation at the International Conference on Turkish Linguistics (ICTL) at Johannes Gutenberg-Universität Mainz.

07/2023 "Japanese gapless relativization: The syntax–prosody interface to semantics." Poster presentation at LFG23: the 28th International Lexical Functional Grammar Conference at the University of Rochester (peer-reviewed).

03/2023 So Miyagawa, Hiroshi Kanayama, Chihiro Taguchi, Nana Tohyama. "沖縄語の Universal Dependencies ツリーバンクコーパスの構築" [Constructing a treebank corpus for Okinawan Universal Dependencies]. Poster presentation at the 29th Annual Meeting of the Association for Natural Language Processing.

03/2023 **Chihiro Taguchi**, So Miyagawa. "形態論情報付き日本語 Universal Dependencies" [Japanese Universal Dependencies with morphological information]. Poster presentation at the 29th Annual Meeting of the Association for Natural Language Processing.

07/2022 "Consistent Grammatical Annotation of Turkic Languages for more universal Universal Dependencies." Oral presentation at the Workshop on Computational Linguistics on East Asian languages.

06/2021 “Mermaid Construction in Tatar.” Oral presentation at the 162nd Meeting of the Linguistic Society of Japan.
11/2020 “Raising to Quirky Subject in Tatar.” Oral presentation at the 161st Meeting of the Linguistic Society of Japan.

Invited talks

10/2026 “Automating recording transcription and Universal Dependencies annotation.” AutoGloss Technical Workshop.
05/2026 “Language Complexity and Speech Recognition Accuracy: Orthographic Complexity Hurts, Phonological Complexity Doesn’t.” Invited talk in the seminar “Advanced Topics in NLP” at Karlsruhe Institute of Technology.
03/2026 “Natural language processing for endangered languages.” Invited lecture in “82-572: Special Topics in Japanese Studies (Japanese Linguistics)” at Carnegie Mellon University.
11/2025 “音声からテキストへ：AI が文字起こしを学習する仕組み” [From speech to text: How AI learns to transcribe]. Invited talk at the Tokyo University of Foreign Studies.
06/2025 “Universal Dependencies Tutorial.” Tutorial given at the Department of Linguistics, University of Tokyo.
03/2025 “人工知能を用いたタタール語談話資料の書き起こしとアノテーション” [AI transcription and annotation of Tatar discourse data]. Talk at the 2024 3rd Research Meeting of the ILCAA Joint Research Project “A study on the grammar of discourse in Eurasian languages.”
03/2025 “少数言語のための音声認識クックブック: キチュア語を事例に” [Automatic speech recognition cookbook for minority languages: A case study of Kichwa]. Invited talk at the 2024 KAKENHI Joint Research Meeting (JP24K00074, JP24K03232, JP24K00061).
09/2024 “Language Complexity and Speech Recognition Accuracy: Orthographic Complexity Hurts, Phonological Complexity Doesn’t.” Invited talk at the NLP Colloquium.
12/2023 “Building an Automatic Speech Recognition Dataset and Model for an Extremely Low-Resource Language.“ Invited talk at the Notre Dame NL+ Seminar.
07/2023 “Bridging Natural Language Processing and Descriptive Linguistics with Universal Dependencies.” Oral presentation at the Field Linguistics Workshop: Grammatical Studies Workshops 23 “Grammatical Studies and Digital Humanities (2)”.
07/2023 “Incorporating AI-based speech transcription into language documentation: A case study of Imbabura Kichwa.” Poster presentation at the LSA Institute.
10/2022 “Cross-linguistic analysis of the Mermaid Constructions in LFG.” Invited talk at the 33rd South of England LFG Meetings.
03/2022 “Mermaid Construction as raising with a nominal predicate.” Invited talk at the 74th Language Lunch At Edinburgh.
10/2021 “Text Processing of Open-Access Jinghpaw Data on Google Colaboratory.” Oral presentation at the Follow-up Meeting on Intensive Language Course 2019 (Jinghpaw) / The 3rd Meeting on Kachin Studies.

Honors and Awards

12/2025 **“Biggest Improvement over Baseline” Winner** in Shared Task: Mozilla Common Voice Spontaneous Speech ASR.
07/2025 **Professionalization Grant**, Kellogg Institute for International Studies, University of Notre Dame.
07/2025 **General Institute Fellowship**, Linguistic Society of America.
09/2024 **The 2024 Lucy Societal Impact Award**, Lucy Family Institute for Data & Society.
08/2024 **Outstanding Paper Award**, Association for Computational Linguistics.
08/2024 **Senior Area Chair’s Award**, Association for Computational Linguistics.
06/2024 **Conference Travel Grant**, Kellogg Institute for International Studies, University of Notre Dame.
05/2024 **Leadership Advancing Socially Engaged Research (LASER) Award**, University of Notre Dame.
05/2023 **Professionalization Grant**, Kellogg Institute for International Studies, University of Notre Dame.
03/2023 **委員特別賞** (Committee’s Honorable Mention), Association for Natural Language Processing.

03/2023 **Summer Language Abroad Grant**, Center for the Study of Languages and Cultures, University of Notre Dame.
09/2021 **JASSO Scholarship for Graduate-level Study Abroad**, Ministry of Education, Culture, Sports, Science and Technology, Japan.
06/2021 **CICP NAIST Multilingual Corpus**, Nara Institute of Science and Technology.
04/2019 **Grand Prix Winner** at the 7th International Olympiad of the Tatar Language and Literature, Ministry of Education and Science, Republic of Tatarstan, Russia.
09/2018 **Exchange Program Scholarship**, Gyomu-Super Japan Dream Foundation
11/2014 **Honorable Mention** at the Keio University Academic Writing Contest, Keio University.

Teaching / Mentoring

04/2026 – 07/2026 **Mentor** for Larry Zhang (BS student) at Carnegie Mellon University.
09/2025 – 12/2025 **Teaching Assistant** for "Natural Language Processing" CSE 40657, University of Notre Dame.

Academic services

10/2026 – Present **Member**, Committee on Linguistic Institutes and Fellowships, Linguistic Society of America.
10/2026 **Reviewer**, 13th International Conference on Information Management and Big Data (SIMBig2026).
10/2026 **Reviewer**, ACL Rolling Review - October 2026.
08/2026 – 06/2026 **Reviewer**, Multilingual Representation Learning (MRL) Workshop.
03/2026 – 05/2026 **Program Committee** (reviewer), SIGUL 2026 Joint Workshop with ELE, EURALI, and DCLRL "Towards Inclusivity and Equality: Language Resources and Technologies for Under-Resourced and Endangered Languages" at LREC 2026.
03/2026 – 05/2026 **Program Committee** (reviewer), DialRes-LREC26.
01/2026 – 07/2026 **Reviewer**, ACL Rolling Review - January 2026.
05/2025 – 11/2025 **Reviewer**, ACL Rolling Review - May 2025.
12/2023 – 05/2024 **Program Committee** (reviewer), 2nd Workshop on Resources and Technologies for Indigenous, Endangered and Lesser-resourced Languages in Eurasia (EURALI) at LREC–COLING 2024.

Grants and projects

02/2026 – Present **Collaborative Researcher** in Grants-in-Aid for Scientific Research 26K00144: **自然言語処理技術を援用した日琉消滅危機言語の談話資料グロス付け自動化に関する研究** [Research on automated interlinear glossing of spoken data for endangered Japonic languages leveraging natural language processing], Japan Society for the Promotion of Science (JSPS). PI: Dr. Michinori Shimoji.
02/2026 – Present **Collaborative Researcher** in Grants-in-Aid for Scientific Research (A) (General) 26H01896: **複雑音韻体系言語の自動音声認識モデル開発と大規模コーパス編纂** [Development of automatic speech recognition models and corpora for languages with a complex phonological system], Japan Society for the Promotion of Science (JSPS). PI: Dr. Hirosi Nakagawa.
01/2026 – Present **Principal Investigator** for the Kellogg Doctoral Research Grant: **The Inheritance of Indigenous Languages: Documentation of Kichwa–Spanish bilingualism in Ecuador**.
04/2025 – Present **Collaborative Researcher** in Grants-in-Aid for Scientific Research (A) (General) 25H00465: **アジア・アフリカのフィールド資料の研究資源化：言語研究の新地平の開拓に向けて** [From field data to research resources for languages of Asia and Africa: Advancing new frontiers in linguistic research], Japan Society for the Promption of Science (JSPS). PI: Dr. Tokusu Kurebito.
07/2024 – 01/2025 **Principal Investigator** for the Kellogg Doctoral Research Grant: **Developing linguistic technologies and digital resources for Kichwa, the minoritized endangered indigenous language of Ecuador**.

01/2024 – Present **Collaborative Researcher** in ILCAA Joint Research Project "A study on the grammar of discourse in Eurasian languages," Research Institute for Languages and Cultures of Asia and Africa, Tokyo University of Foreign Studies. PI: Dr. Norikazu Kogura.

08/2022 – 07/2025 **Graduate Research Assistant** for **NEH DLI-DEL Collaborative Research: Language Documentation with an Artificial Intelligence Helper**, Award No. 2109709. PI: Dr. Antonios Anastasopoulos.

Skills

Programming; OS: Python, R, C++, Shell script; Mac, Linux, Windows

Software/libraries: PyTorch, Tensorflow, Flask, Git, AWS, Docker, ELAN, Praat

Languages:

Native:	Japanese
Fluent (C2):	English
Advanced (C1):	Spanish
Intermediate (B2):	Korean, Mandarin Chinese, Russian, French, German, Tatar
Research:	Ecuadorian Highland Kichwa, Jinghpaw, Ikema Miyakoan

Other academic affilations

Kellogg Institute Doctoral Student Affiliate
Association of Computational Linguistics (ACL)
International Speech Communication Association (ISCA)
Linguistic Society of America (LSA)
言語処理学会 [Association for Natural Language Processing]